Week 4 Update

Michael Fang

April 2026

Week 4 Update

Week 4 Update
Michael Fang
April 2026

Short update this week. Main topics:

- Why α can't exceed 3
- What controls the $\bar{\Lambda}$ ranking
- Metallic-mean tiles and $\bar{\Lambda}$ limit (extended to $n = 500$)
- Yuan-Torquato improved spreadability
- 138 $|\det M|$=2 chains

Recap

Last week: proved $2<\alpha<3$ gap for $|\det M|=1$, found first $|\det M|=2$ example at $\alpha = 2.071$, Silver $\bar{\Lambda} \approx 1/4$, metallic-mean $\bar{\Lambda}$ approaching $\sim 1/3$.

Since last meeting:

- Worked out why α can't exceed 3 for $|\det M|=1$ substitutions
- Understood what determines the $\bar{\Lambda}$ ranking
- Generated metallic-mean tile visualizations ($n = 1$ through 10)
- Extended metallic-mean $\bar{\Lambda}$ to $n = 500$
- Implemented Yuan-Torquato (2026) improved spreadability
- Computed $\bar{\Lambda}$ for all 138 $|\det M|=2$ chains

Recap

Last week: proved $2<\alpha<3$ gap for $|\det M|=1$, found first $|\det M|=2$ example at $\alpha = 2.071$, Silver $\bar{\Lambda} \approx 1/4$, metallic-mean $\bar{\Lambda}$ approaching $\sim 1/3$.

Since last meeting:

- Worked out why α can't exceed 3 for $|\det M|=1$ substitutions
- Understood what determines the $\bar{\Lambda}$ ranking
- Generated metallic-mean tile visualizations ($n = 1$ through 10)
- Extended metallic-mean $\bar{\Lambda}$ to $n = 500$
- Implemented Yuan-Torquato (2026) improved spreadability
- Computed $\bar{\Lambda}$ for all 138 $|\det M|=2$ chains

Why α can't exceed 3

For a rank-r substitution matrix with $|\det M| = 1$ and all $|\lambda_i| < 1$ for $i \geq 2$:

$$\underbrace{\lambda_1 \prod_{i=2}^{r} |\lambda_i|}_{=|\det M|=1} \;\leq\; \lambda_1 \, |\lambda_2|^{r-1} \quad \Rightarrow \quad |\lambda_2| \;\geq\; \lambda_1^{-1/(r-1)}$$

Substituting into $\alpha = 1 - 2\ln|\lambda_2| / \ln \lambda_1$:

$$\boxed{\alpha \;\leq\; 1 + \frac{2}{r-1}}$$

Rank r	$\alpha_{\max}$	Example
2	**3**	all metallic means (Fibonacci, Silver, ...)
3	2	504 cubic matrices with complex pair
4	5/3	
$r \to \infty$	$\to 1^+$	

Why α can't exceed 3
For a rank-r substitution matrix with $|\det M| = 1$ and all $|\lambda_i| < 1$ for $i \geq 2$:
$$\underbrace{\lambda_1 \prod_{i=2}^{r} |\lambda_i|}_{=|\det M|=1} \leq \lambda_1 |\lambda_2|^{r-1} \quad \Rightarrow \quad |\lambda_2| \geq \lambda_1^{-1/(r-1)}$$
Substituting into $\alpha = 1 - 2\ln|\lambda_2| / \ln \lambda_1$:
$$\boxed{\alpha \leq 1 + \frac{2}{r-1}}$$
Rank r | $\alpha_{\max}$ | Example
2 | **3** | all metallic means (Fibonacci, Silver, ...)
3 | 2 | 504 cubic matrices with complex pair
4 | 5/3
$r \to \infty$ | $\to 1^+$

The bound is tight for rank 2: metallic means saturate it. For rank 3: saturated by the complex-pair case ($\alpha = 2$ exactly).
Higher rank $\to$ more eigenvalues sharing the determinant constraint $\to$ $|\lambda_2|$ forced larger $\to$ α forced smaller.
For $|\det M| > 1$, the bound generalizes to $\alpha \leq 1 + \frac{2}{r-1}\left(1 - \frac{\ln|\det M|}{\ln \lambda_1}\right)$.

What determines the $\bar{\Lambda}$ ranking?

$\bar{\Lambda}$ is smallest for patterns closest to perfect lattice order.

- **Lattice**: $\bar{\Lambda} = 1/6$, the minimum benchmark. No spacing disorder, so there is no residual deviation from lattice order.
- **Metallic means**: as n grows, short tiles become asymptotically rare, but the sequence still contains two spacings. Numerically, $\bar{\Lambda}$ increases with n over the range studied.
- **URL**: $\bar{\Lambda} = (1 + a^2)/6$ exactly. Larger displacement amplitude a means larger departures from the parent lattice, so $\bar{\Lambda}$ increases.

Interpretation: within Class I, larger $\bar{\Lambda}$ reflects stronger or more frequent departures from perfect lattice order.

What determines the $\bar{\Lambda}$ ranking?

$\bar{\Lambda}$ is smallest for patterns closest to perfect lattice order.

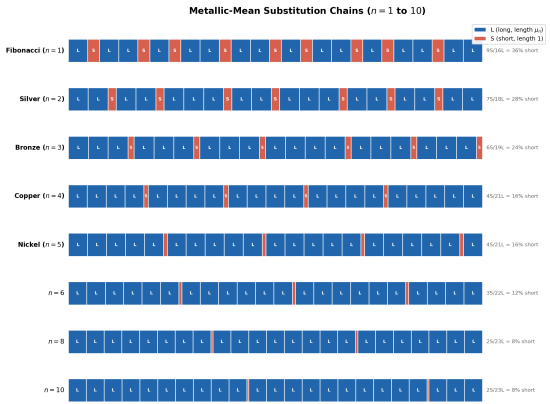
- **Lattice**: $\bar{\Lambda} = 1/6$, the minimum benchmark. No spacing disorder, so there is no residual deviation from lattice order.
- **Metallic means**: as n grows, short tiles become asymptotically rare, but the sequence still contains two spacings. Numerically, $\bar{\Lambda}$ increases with n over the range studied.
- **URL**: $\bar{\Lambda} = (1 + a^2)/6$ exactly. Larger displacement amplitude a means larger departures from the parent lattice, so $\bar{\Lambda}$ increases.

Interpretation: within Class I, larger $\bar{\Lambda}$ reflects stronger or more frequent departures from perfect lattice order.

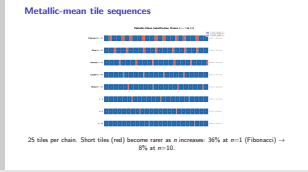
Metallic n	$\bar{\Lambda}$
1	0.199
2	0.250
3	0.281
5	0.303
10	0.326
20	0.332
50	0.333
100	0.334

Reference	α	$\bar{\Lambda}$
Integer lattice	—	0.167
URL ($a = 0$)	2	0.167
URL ($a = 0.1$)	2	0.168
URL ($a = 0.3$)	2	0.182
URL ($a = 0.5$)	2	0.208
$\lvert\det M\rvert = 2$ family	2.07–2.39	0.32–5.08
URL ($a = 1$)	2	0.333
Bombieri-Taylor	1.545	0.377

Metallic-mean tile sequences



25 tiles per chain. Short tiles (red) become rarer as n increases: 36% at n=1 (Fibonacci) → 8% at n=10.



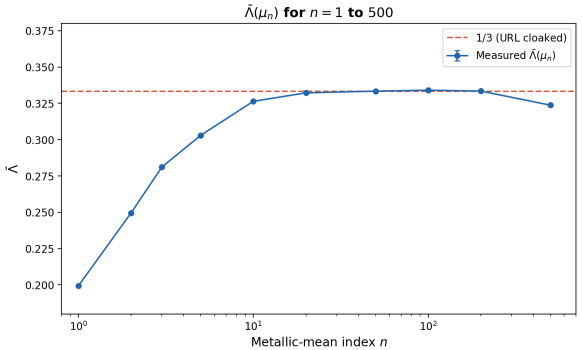
Rules: $S \to L$, $L \to L^n S$.

$n = 1$ (Fibonacci): $S \to L$, $L \to LS$. Balanced quasi-periodic mix.

$n = 10$: $L \to LLLLLLLLLS$. Almost all L tiles, rare S defects.

S fraction $= 1/\mu_n \to 0$ as $n \to \infty$. That asymptotic rarity is the safe statement.

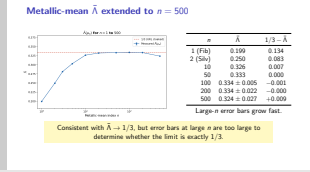
Metallic-mean $\bar{\Lambda}$ extended to $n = 500$



n	$\bar{\Lambda}$	$1/3 - \bar{\Lambda}$
1 (Fib)	0.199	0.134
2 (Silv)	0.250	0.083
10	0.326	0.007
50	0.333	0.000
100	0.334 ± 0.005	-0.001
200	0.334 ± 0.022	-0.000
500	0.324 ± 0.027	$+0.009$

Large-n error bars grow fast.

Consistent with $\bar{\Lambda} \to 1/3$, but error bars at large n are too large to determine whether the limit is exactly $1/3$.

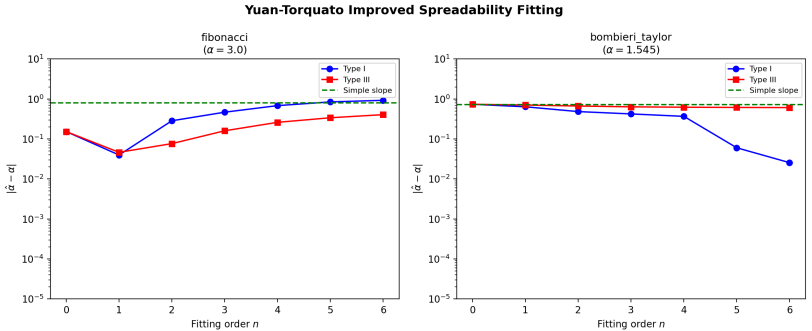


$n = 1 \ldots 20$: monotone increasing toward $1/3$ from below. $n = 50$: 0.333, $n = 100$: 0.334 — right at $1/3$. $n = 200$: 0.334 ± 0.022 — consistent but huge error bar. $n = 500$: 0.324 ± 0.027 — actually below $1/3$, but noise dominates.

At large n, chains grow by factor $\sim n$ per iteration. $n = 500$ with 3 iterations gives 125M tiles, but the quasi-periodic structure is so coarse (only 3 "generations") that variance estimation is very imprecise.

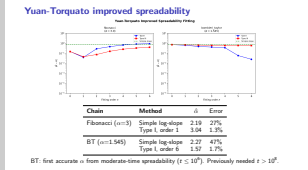
Bottom line: data is consistent with $1/3$ as the limit, but we can't rule out a slightly different value.

Yuan-Torquato improved spreadability



Chain	Method	$\hat{\alpha}$	Error
Fibonacci (α=3)	Simple log-slope	2.19	27%
	Type I, order 1	3.04	1.3%
BT (α=1.545)	Simple log-slope	2.27	47%
	Type I, order 6	1.57	1.7%

BT: first accurate α from moderate-time spreadability ($t \leq 10^6$). Previously needed $t > 10^8$.



Main project result: BT becomes accurately measurable from moderate-time spreadability ($t \leq 10^6$), instead of requiring $t > 10^8$.

BT needs high order because $|\lambda_2| = 0.802$ gives unusually slow decay of correction terms.

Fibonacci is already well captured at Type I, order 1; higher order does not improve the fit in a controlled way.

Useful for difficult cases like BT, but not a general replacement for variance-based extraction.

All $|\det M|$=2 chains with $\alpha \in (2,3)$

Searched all 2×2 matrices with $|\det M| = 2$, $|\lambda_2| < 1$, row sums ≤ 10:

α	Chains	$\bar{\Lambda}$ range
2.071	12	[0.32, 2.01]
2.087	10	[0.34, 1.57]
2.175	18	[0.32, 3.36]
2.271	16	[0.34, 3.76]
2.301	18	[0.33, 4.05]
2.362	12	[0.34, 4.80]
2.393	4	[0.34, 5.08]

138 chains total
12 distinct α values
(7 shown)

$\alpha \in [2.07, 2.39]$

$\bar{\Lambda}$ varies by factors of
5–15 *within each α group*

α alone does **not**
determine $\bar{\Lambda}$.

All $|\det M|$=2 chains with $\alpha \in (2,3)$
Searched all 2×2 matrices with $|\det M| = 2$, $|\lambda_2| < 1$, row sums ≤ 10:
138 chains total
12 distinct α values
(7 shown)
$\alpha \in [2.07, 2.39]$
$\bar{\Lambda}$ varies by factors of
5–15 *within each α group*
α alone does **not**
determine $\bar{\Lambda}$.

Expanded from original 28 matrices (row sums ≤ 6) to 138 (row sums ≤ 10).
Formula: $\alpha = 3 - 2\ln|\det M|/\ln\lambda_1$. With $|\det M| = 2$ and $\lambda_1 > 4$: $\alpha \in (2.0, 2.4)$.
Key finding: two matrices with identical eigenvalues can have $\bar{\Lambda}$ differing by a factor of 15. This means α captures only the leading spectral behavior, while $\bar{\Lambda}$ is sensitive to the full spatial arrangement of tiles.
Smallest $\bar{\Lambda} \approx 0.32$–$0.34$ across all groups, close to $1/3$ (URL cloaked value).

Summary

Since last meeting:

- $\alpha \leq 1 + 2/(r-1)$ for rank-r matrices with $|\det M|=1$; $\alpha=3$ is the global maximum (rank 2 only)
- $\bar{\Lambda}$ ranking: controlled by tile disparity, displacement amplitude, spectral constraint strength
- Metallic-mean tiles visualized ($n=1\ldots10$); $\bar{\Lambda}$ extended to $n=500$ (consistent with $\sim1/3$, noisy at large n)
- Yuan-Torquato: BT α error reduced from 47% to 1.7%
- 138 $|\det M|=2$ chains computed; α does not determine $\bar{\Lambda}$

Possible next steps:

- Write JP results sections
- Higher-precision metallic-mean runs at large n
- $|\det M| = 3$ chains to reach α closer to 3?

Summary
Since last meeting:
- $\alpha \leq 1 + 2/(r-1)$ for rank-r matrices with $|\det M|=1$; $\alpha=3$ is the global maximum (rank 2 only)
- $\bar{\Lambda}$ ranking: controlled by tile disparity, displacement amplitude, spectral constraint strength
- Metallic-mean tiles visualized ($n=1\ldots10$); $\bar{\Lambda}$ extended to $n=500$ (consistent with $\sim1/3$, noisy at large n)
- Yuan-Torquato: BT α error reduced from 47% to 1.7%
- 138 $|\det M|=2$ chains computed; α does not determine $\bar{\Lambda}$
Possible next steps:
- Write JP results sections
- Higher-precision metallic-mean runs at large n
- $|\det M| = 3$ chains to reach α closer to 3?

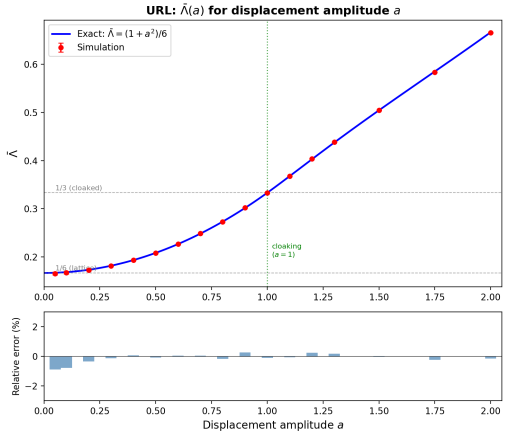
References

- Torquato & Stillinger (2003), Phys. Rev. E **68**, 041113
- Oğuz et al. (2019), Acta Cryst. A **75**, 3–13
- Klatt, Kim & Torquato (2020), Phys. Rev. E **101**, 032118
- Torquato (2021), Phys. Rev. E **104**, 054102
- Yuan & Torquato (2026), arXiv:2602.17873
- Hitin-Bialus et al. (2024), Phys. Rev. E **109**, 064108

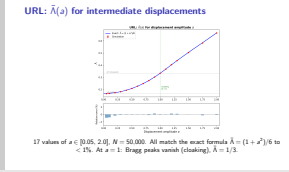
References

- Torquato & Stillinger (2003), Phys. Rev. E **68**, 041113
- Oğuz et al. (2019), Acta Cryst. A **75**, 3–13
- Klatt, Kim & Torquato (2020), Phys. Rev. E **101**, 032118
- Torquato (2021), Phys. Rev. E **104**, 054102
- Yuan & Torquato (2026), arXiv:2602.17873
- Hitin-Bialus et al. (2024), Phys. Rev. E **109**, 064108

URL: $\bar{\Lambda}(a)$ for intermediate displacements



17 values of $a \in [0.05, 2.0]$, $N = 50{,}000$. All match the exact formula $\bar{\Lambda} = (1 + a^2)/6$ to $< 1\%$. At $a = 1$: Bragg peaks vanish (cloaking), $\bar{\Lambda} = 1/3$.



Cloaking: the displacement form factor $\tilde{f}(k) = \sin(\pi ka)/(\pi ka)$ vanishes at all non-zero reciprocal lattice vectors when a is an integer.

Klatt et al. (2020) derived the formula analytically.

g_2-invariant patterns

From Torquato & Stillinger (2003): a g_2-invariant process has a fixed pair correlation $g_2(r)$ across a range of densities. At terminal density: $S(0) = 0$ (hyperuniform).

Pattern	Construction	$\bar{\Lambda}$
Step-function g_2	hard rods at close packing	1/6 (lattice)
Lattice-gas	one point uniform per cell	1/3 (= URL a=1)
Gauss. shuffled, σ=0.1	$\mathcal{N}(0, 0.01)$ per site	0.20 ± 0.02
Gauss. shuffled, σ=0.2	$\mathcal{N}(0, 0.04)$ per site	0.250 ± 0.007
Gauss. shuffled, σ=0.3	$\mathcal{N}(0, 0.09)$ per site	0.342 ± 0.001

The lattice-gas is just URL a=1 with a phase offset. Gaussian shuffled lattice at σ=0.1 matches Fibonacci; σ=0.2 matches Silver (1/4).

g_2-invariant patterns

From Torquato & Stillinger (2003): a g_2-invariant process has a fixed pair correlation $g_2(r)$ across a range of densities. At terminal density: $S(0) = 0$ (hyperuniform).

Pattern	Construction	$\bar{\Lambda}$
Step-function g_2	hard rods at close packing	1/6 (lattice)
Lattice-gas	one point uniform per cell	1/3 (= URL a=1)
Gauss. shuffled, σ=0.1	$\mathcal{N}(0, 0.01)$ per site	0.20 ± 0.02
Gauss. shuffled, σ=0.2	$\mathcal{N}(0, 0.04)$ per site	0.250 ± 0.007
Gauss. shuffled, σ=0.3	$\mathcal{N}(0, 0.09)$ per site	0.342 ± 0.001

The lattice-gas is just URL a=1 with a phase offset. Gaussian shuffled lattice at σ=0.1 matches Fibonacci; σ=0.2 matches Silver (1/4).

The 1D g_2-invariant processes all reduce to perturbed lattices. "Lattice-gas": tessellate line into unit cells, place one point uniformly in each. This is $j + U(0, 1)$ vs. URL's $j + U(-1/2, 1/2)$.